

Lesson 7 — Button & Tap Triggers

Matches LEGO Teacher Portal lesson: "Buzz and Bloom".

Objective (I can...): I can make my model respond to a button press or a screen tap. **CS concept:** Input events from a control · **Builds on:** Lesson 6 (events as start signals).

AI Connection (2 min): Tapping a smart speaker or saying "Hey" is an input event that wakes the AI. *First-principles question:* "Why might a device wait for a signal instead of always listening?"

Materials: LEGO CS & AI kit + device per pair. **Model reuse:** Events model from Lesson 6 — code only.

60-minute flow - Warm-up (8): Human buttons — one student is a "button," another the "machine" that acts only when pressed. Reinforce input → action. - **Concept + demo (10):** Demo two event blocks (physical press and screen tap) driving two different actions. - **Build/Code (30):** On the built model, pairs program two triggers — one press event, one tap event — each starting a different short action. *Swap at 15 min.* - **Challenge (7):** Make one press do a two-step action (move then light) while a tap does something else. - **Share + pack-up (5):** Pairs demo both triggers and explain which input does what. Model stays built.

Pair roles: Driver codes trigger 1, Navigator plans trigger 2; swap at 15 min so each builds one. (*Trio: third owns testing both.*)

Differentiation: *Support* — one trigger only, with a twin example to adapt. *Extension / fast-finisher* — add a third distinct trigger. *Reteach* — do one trigger end-to-end together, then let them mirror it for the second.

Exit check: Pair shows two different inputs producing two different actions reliably. **At-home (optional):** Design a paper remote; label what each button triggers. ~10 min.
